

MUHAMMAD USMAN

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EDUCATION

Sukkur IBA University
Bachelor's of Engineering in Computer Systems Engineering

Aug 2022 - May 2026
CGPA: 3.53/4.00

Intermediate
B.I.S.E Sukkur

Aug 2020 - May 2022
Grade: A1

WORK EXPERIENCE

10Pearls – Data Science Intern (Remote)

Jul 2026- Sept 2026

- Built a 100% serverless end-to-end MLOps pipeline using GCP/Vertex AI and Hopsworks to forecast 3-day AQI, with 24× daily automated feature ingestion from AQICN/OpenWeather APIs and daily CI/CD model retraining via GitHub Actions.
- Benchmarked 3 ML models, Random Forest, Ridge Regression, and TensorFlow using RMSE, MAE, and R^2 , and developed a real-time HTML/CSS/JavaScript dashboard with SHAP for explainable AQI forecasts and hazardous-air-quality alerts.

DevelopersHub Corporation – AI/ML Intern (Remote)

Jun 2025- Aug 2025

- Built and deployed 6+ AI/ML solutions, including LLM chatbots, regression models, and a customer-churn predictor tuned with GridSearchCV, achieving 80–88% accuracy across tasks.
- Implemented a RAG pipeline with LangChain and FAISS and built NLP auto-tagging models that improved support-ticket labeling accuracy by ~30% in zero-shot and few-shot setups.

CodeGradients – Python Intern (On-site)

Jun 2025- Jul 2025

- Developed Python automation and data-processing scripts that cut manual workload by 30–40% and improved workflow reliability.
- Delivered production-ready modules by applying clean coding, structured debugging, and Git workflows, reducing code issues by over 25%.

PROJECTS

- Pearls AQI Predictor** - Built a 100% serverless, end-to-end MLOps pipeline on GCP/Vertex AI and Hopsworks to forecast 3-day Air Quality Index, with automated hourly feature ingestion from AQICN/OpenWeather APIs and an automated CI/CD pipeline (GitHub Actions) for daily model retraining. [View](#)
 - SafeDrive AI (Final Year Project)** - Built a real-time 5-class edge vision system using YOLOv8ncls, NCNN, OpenCV, and Raspberry Pi 4, enabling 100% offline real-time driver anomaly detection with optimized ARM inference and GradCAM-based model explainability. [View](#)
 - Context-Aware Chatbot (RAG)** - Developed a RAG system using LangChain + FAISS enabling semantic retrieval over 500–1000 chunks for context-rich answers. [View](#)
 - Chronos AI** - Developed a dual-memory RAG system (ChromaDB + Mistral-7B) for long-term context retention, boosting response personalization by ~45% across multi-session chats. [View](#)
 - Ascend AI** - Integrated Llama 3, Qwen, and Gemma via a two-stage Flask pipeline for automated roadmapping; achieved ~99.9% uptime and 40% faster profile-to-summary generation. [View](#)
 - COVID-19 CT Scan Classification** - Developed a VGG16 transfer learning model in TensorFlow/Keras on 4,962 CT scans, achieving 96.48% accuracy, 0.97 F1-score, and +1.52% over the baseline. [View](#)
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PUBLICATIONS

5th ICoMET 2026

Evaluating Deep Learning Models for Prohibited Item Detection in Airport X-ray Images [View](#)

- Developed and benchmarked YOLOv8n, Faster R-CNN, and EfficientDet-D0 on the 45K+ image HiXRay dataset, achieving 0.803 mAP@0.5 and 88.41 FPS with YOLOv8n.
 - Built the preprocessing and evaluation pipeline using PyTorch, analyzing model performance with mAP, Precision, Recall, F1-score, and FPS.
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PROFESSIONAL CERTIFICATIONS

GenAI App Development
(PEC, PakAngels, Aspire Pakistan)

Co-Trainer | Gen AI Workshop
(CSE Society, Sukkur IBA)

Quantum Computing
(LinkedIn Learning)

Prompt Engineering
(LinkedIn Learning)

ADDITIONAL

- **Technical Skills:** TensorFlow, Scikit-learn, PyTorch, Hugging Face, LangChain, Git, GitHub, CI/CD (GitHub Actions), AWS, Azure, GCP (Vertex AI), MySQL, NumPy, Pandas, Seaborn, OpenCV, Remote-XY, Arduino IDE, NI Multisim, Jupyter Notebook
- **Programming Languages:** Python, C++, C, JavaScript, Java, Embedded C, MATLAB